

기술 면접 PT

Computer Vision Data Engineer Internship(Robotics)

지원자: 김도원

2025.12.05



김 도원

Undergraduate Researcher

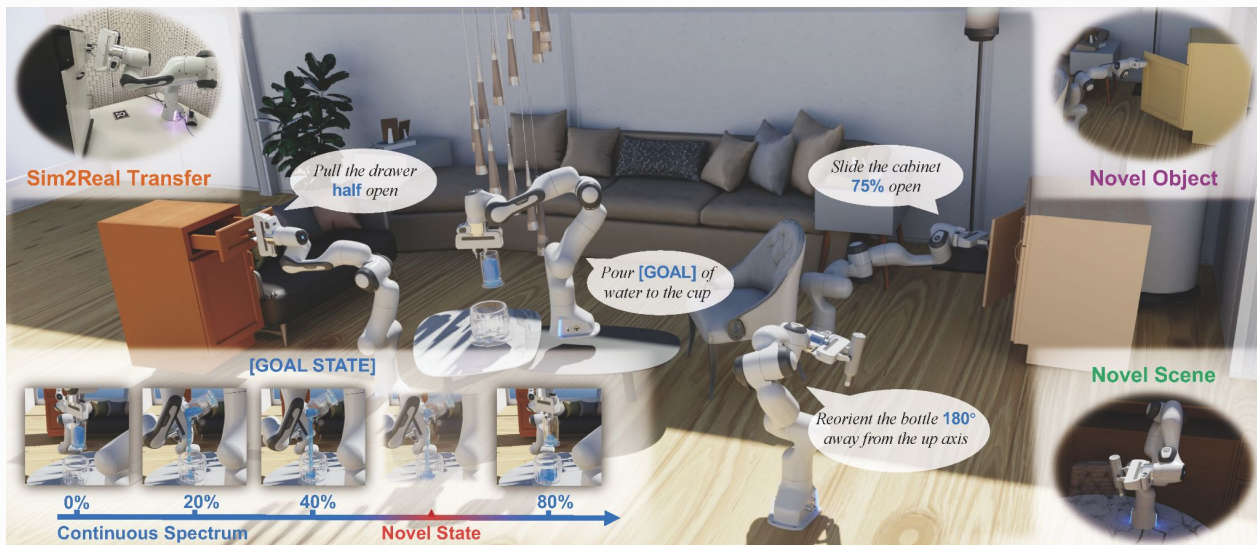
- Field: Physical AI, Robotic data generation, VLA.
- Awards: 1st Place, ARNOLD Challenge.
- GPA: 4.3 / 4.5

“로봇이 다양한 도메인에서 **눈으로 세상을 새롭게 해석**하고 행동할 수 있도록 경험(데이터)을 전해주고 싶습니다.”

2. ARNOLD Challenge (@CVPR 2025)

Robotic Manipulation in Embodied AI

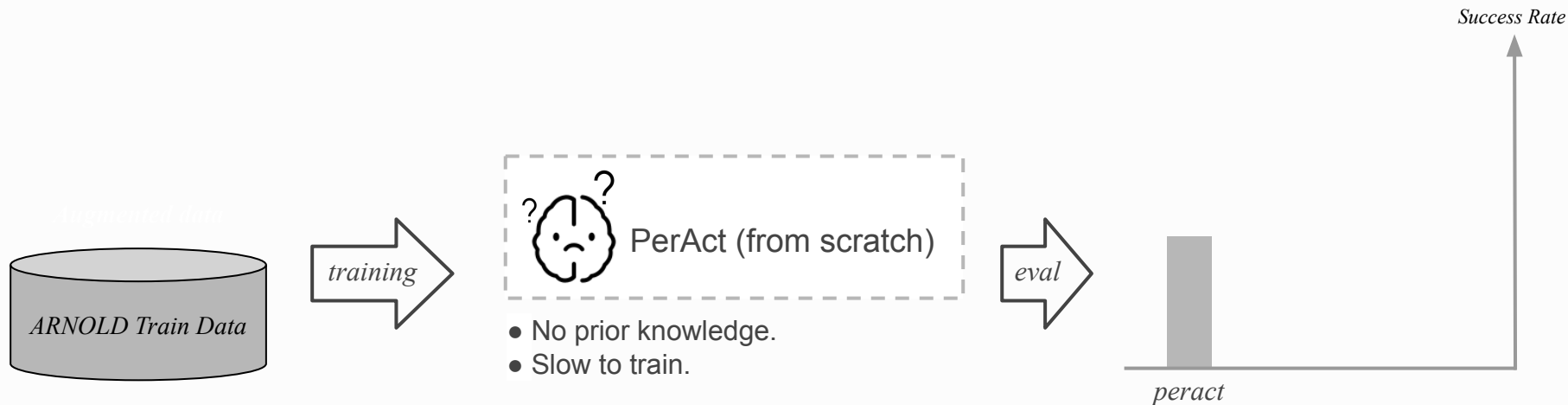
- Goal: **Perceive** and **manipulate** objects to complete designated tasks.
- Challenge: Real-world manipulation requires **generalization** to unseen scenarios.



2. ARNOLD Challenge (@CVPR 2025)

(1) VLA foundation model. ← “Generalization”

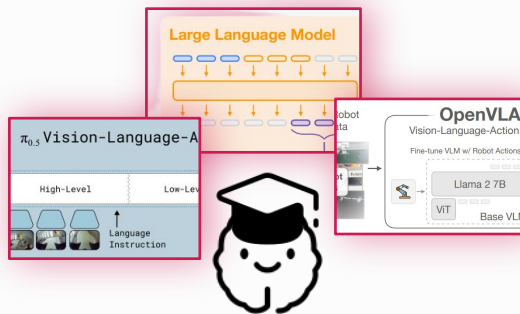
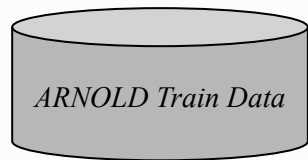
(2) Data augmentation.



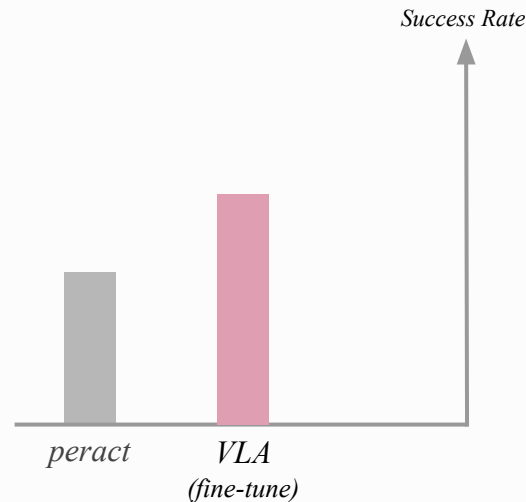
2. ARNOLD Challenge (@CVPR 2025)

(1) VLA foundation model.

(2) Data augmentation.



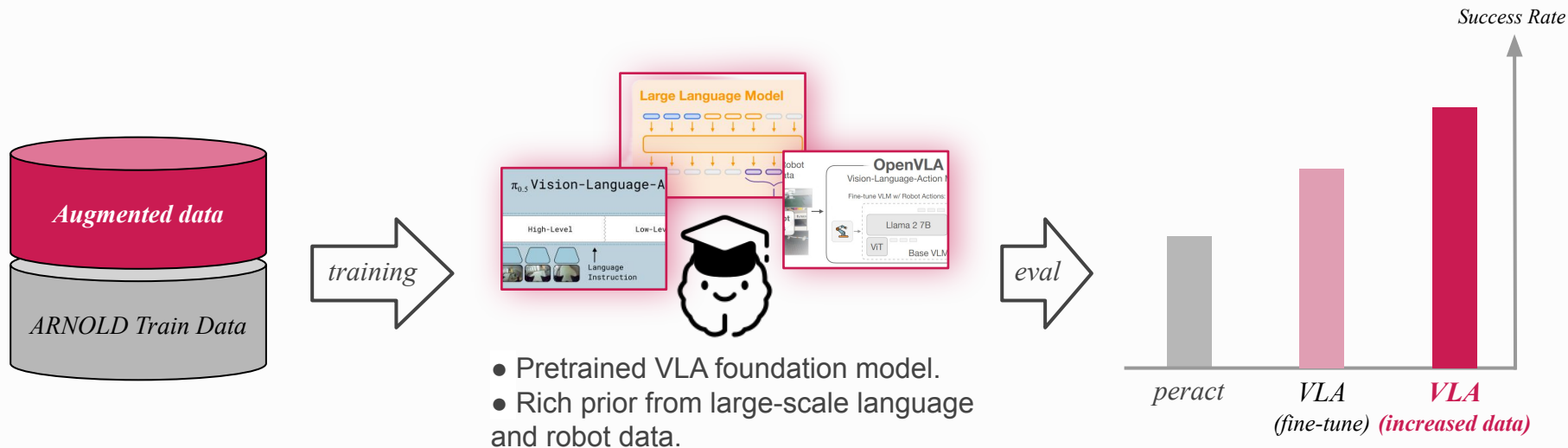
- Pretrained VLA foundation model.
- Rich prior from large-scale language and robot data.



2. ARNOLD Challenge (@CVPR 2025)

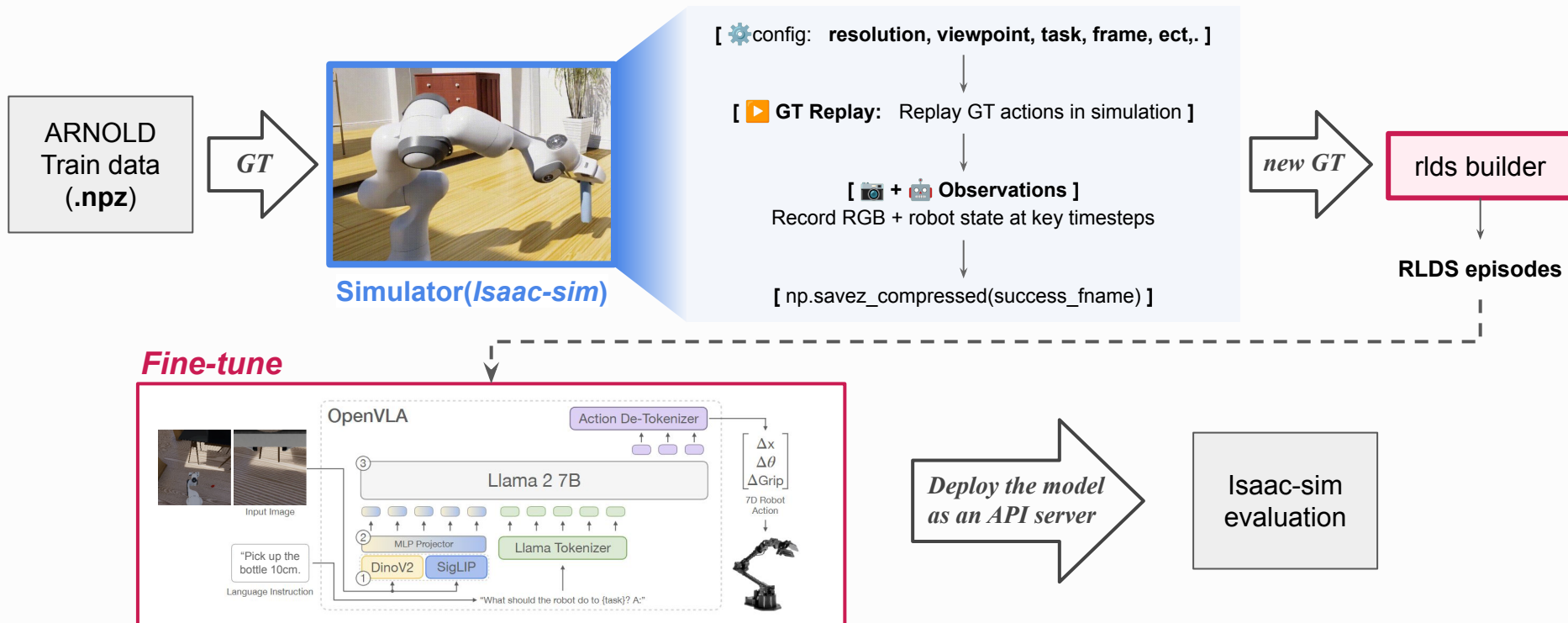
(1) VLA foundation model.

(2) Data augmentation.



2. ARNOLD Challenge (@CVPR 2025)

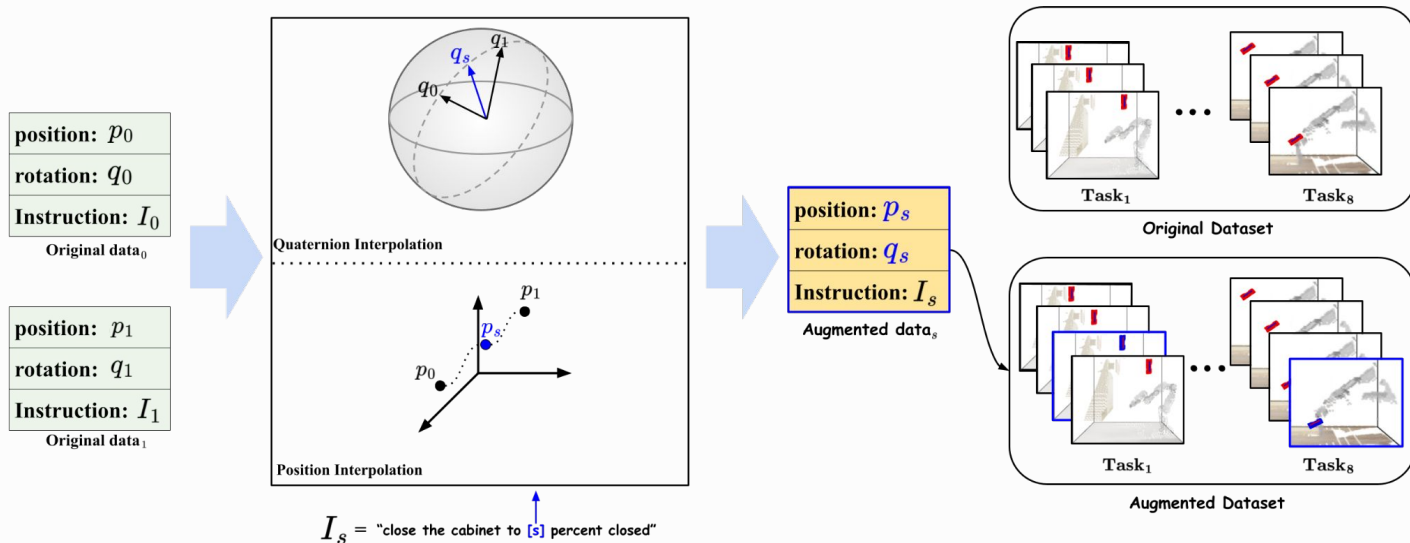
1. [Fine-tuning VLA foundation models (openVLA)]



2. ARNOLD Challenge (@CVPR 2025)

2. [Data augmentation]

1) State Interpolation



2) Dynscene

: (CVPR 2025) Robotic manipulation scene generation method enabling scalable dataset creation.

2. ARNOLD Challenge (@CVPR 2025)

The performance results were as follows, and we won the final first place in the challenge.

		** Success Rate			
		Method			
		Baseline	Ours (w/o SI/DS)	Ours (w/ SI)	Ours
setting	Backbone	PerAct	OpenVLA	OpenVLA	OpenVLA
	State Interpolation	✗	✗	✓	✓
	DynScene	✗	✗	✗	✓
scenario	Test	42.47	53.13	58.75	59.38
	Novel Object	22.84	27.92	31.88	33.33
	Novel Scene	29.73	37.71	41.67	44.58
	Novel State	4.49	26.88	36.46	40.21
	Any State	16.36	33.33	45.63	44.17



3. Follow-up

Some tasks remained hard even after the ARNOLD challenge → we analyzed failure cases.

- If the first grasp is wrong or the scene changes, the agent has no chance to recover.



: a false prediction



: object fall



“Open the drawer completely”

→ **Limitation of the ARNOLD 2-stage protocol.** With a 2-stage protocol, the policy must predict an accurate grasp from limited initial observations.

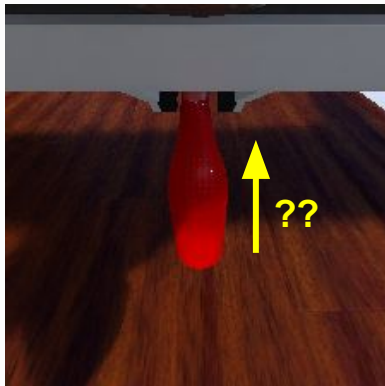
3. Follow-up

Modified Closed-Loop Evaluation

- Rewrite the evaluation to support closed-loop.
- Regenerate training data and train the policy under closed-loop setting.



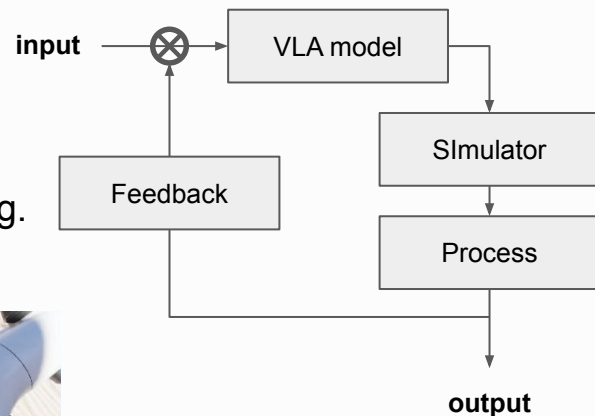
: evaluation



: "Pick up the bottle 10cm"



2 step

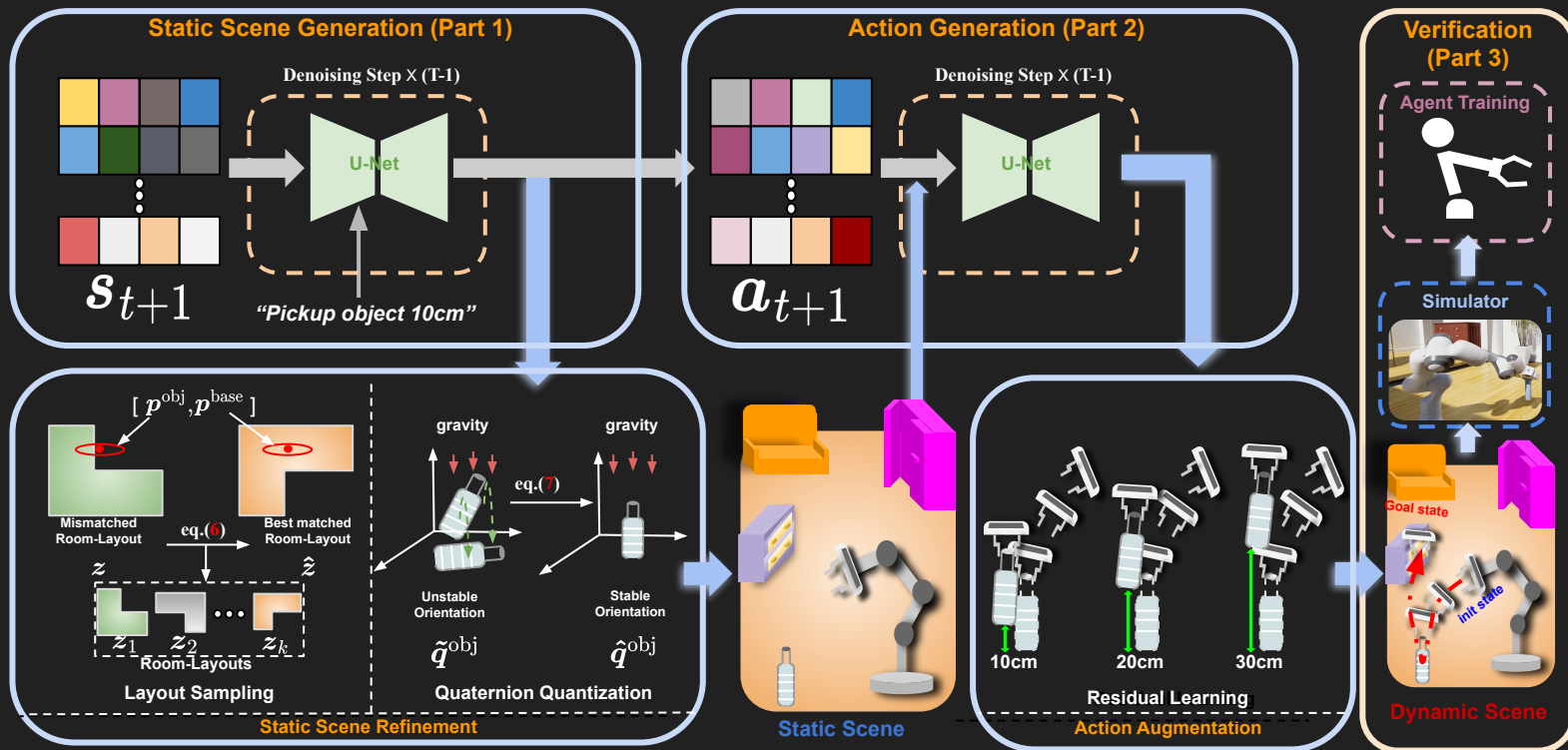


→ VLA Guidance Method Research.

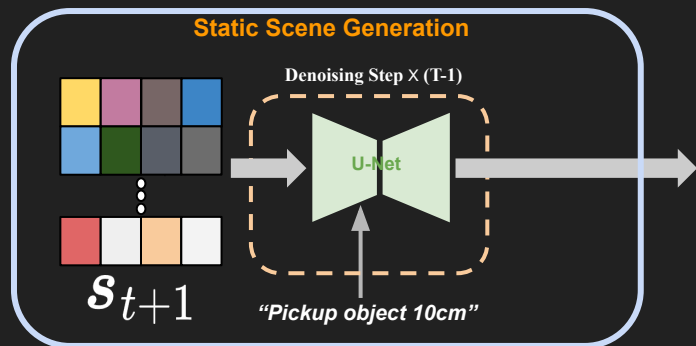
“이러한 경험을 바탕으로 해서, 로봇의 눈과 경험을 기록하고 분석하는 데이터 엔지니어로서
LG AI 연구원 Vision Lab에서 기여할 수 있도록 노력하겠습니다.”

감사합니다.

Framework Overview



- Diffusion model effectively generates **realistic** static scenes aligned with instructions.
- However, requires refinement for **physical plausibility** and **task accessibility**.



Unable to reach the target



Robot-Layout collision

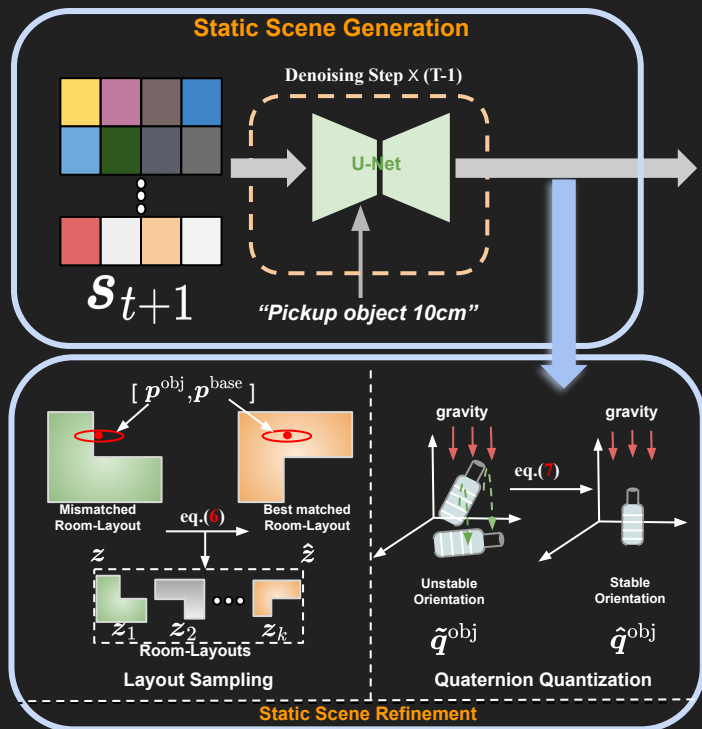


Unstable Orientation



Unintended Opening

- Diffusion model effectively generates **realistic** static scenes aligned with instructions.
- However, requires refinement for **physical plausibility** and **task accessibility**.



- **Layout Sampling** (eq.6)

$$\hat{r} = \arg \min_r (\|p_r^{obj} - \tilde{p}^{obj}\|^2 + \|p_r^{base} - \tilde{p}^{base}\|^2)$$

Unable to reach the target

Robot-Layout collision

Gravity

- **Quaternion Quantization** (eq.7)

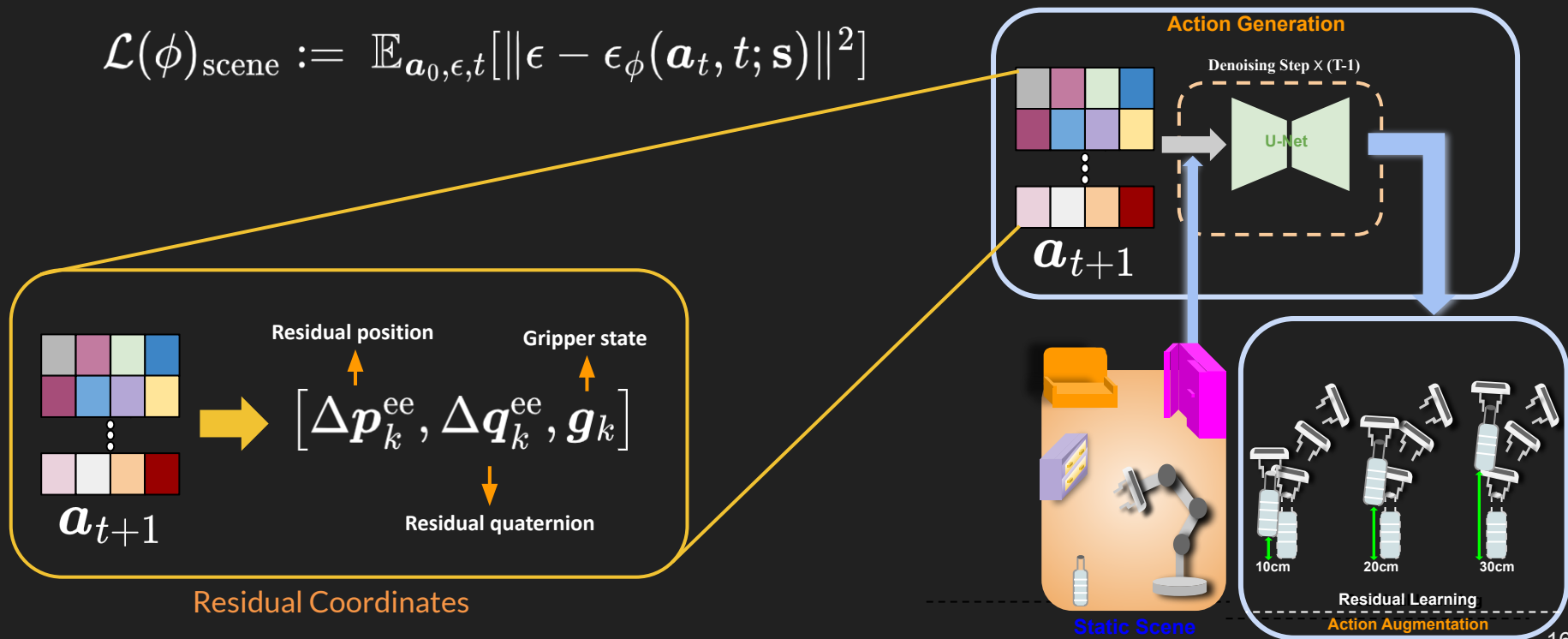
$$\hat{q}^{obj} = \text{round} \left(\frac{\tilde{q}^{obj}}{\delta} \right) \cdot \delta$$

Unstable Orientation

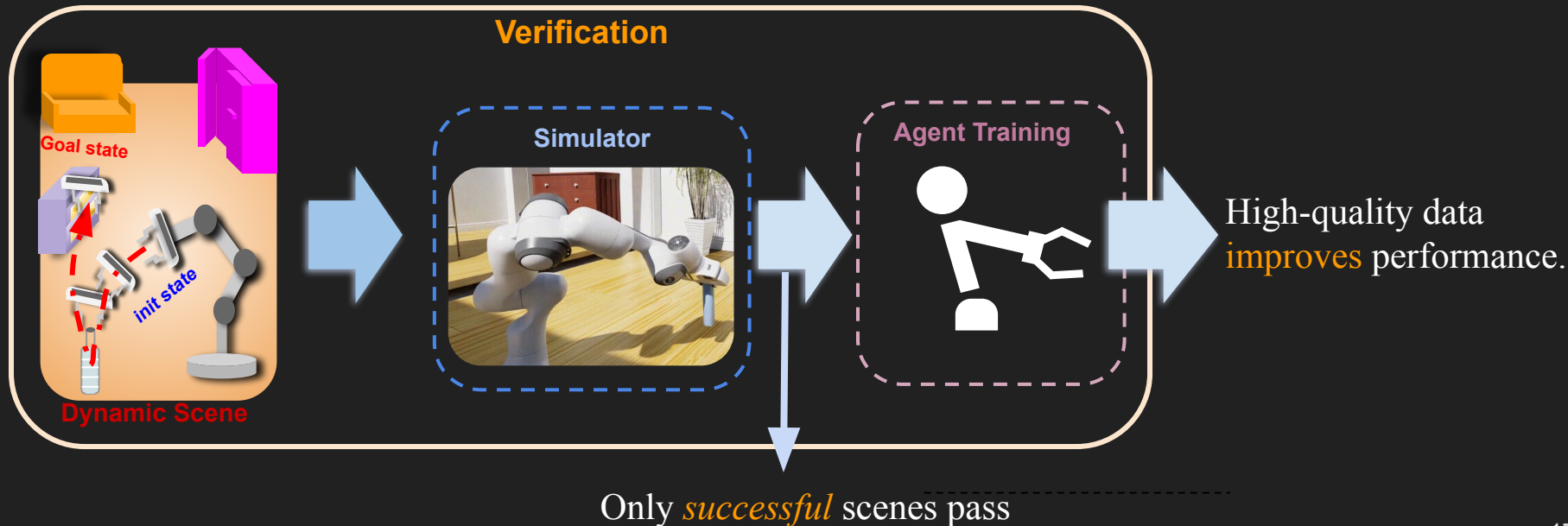
Unintended Opening

- Action generation uses diffusion model conditioned on static scenes.

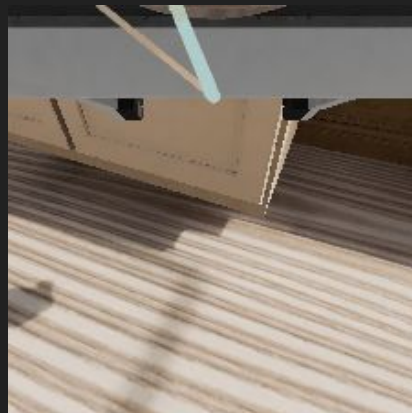
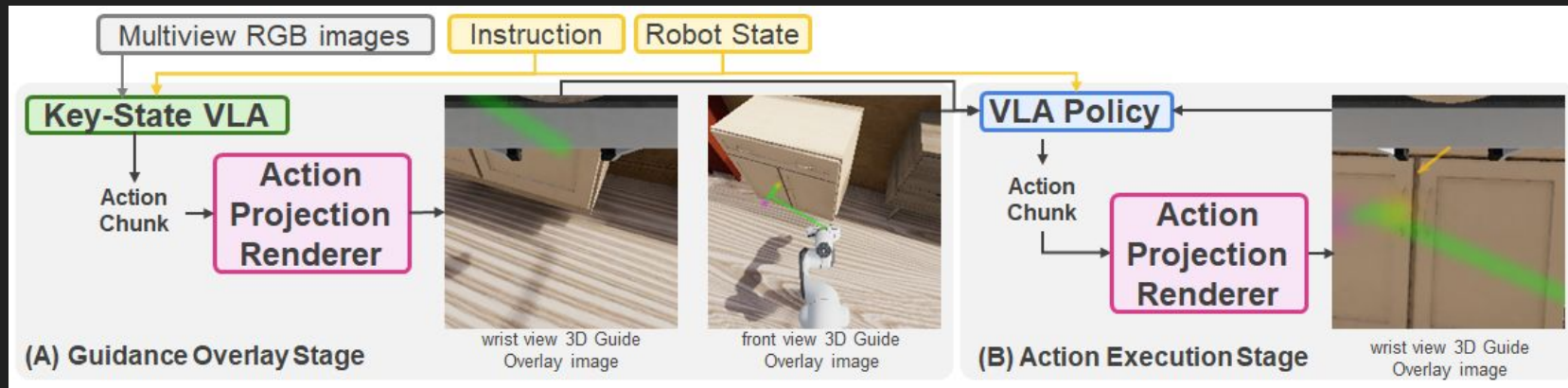
$$\mathcal{L}(\phi)_{\text{scene}} := \mathbb{E}_{\mathbf{a}_0, \epsilon, t} [\|\epsilon - \epsilon_\phi(\mathbf{a}_t, t; \mathbf{s})\|^2]$$



- Inaccurate actions can degrade agent performance and cause task **failures**.
 - Generated dynamic scenes undergo physics simulation verification in *NVIDIA Isaac Sim*.

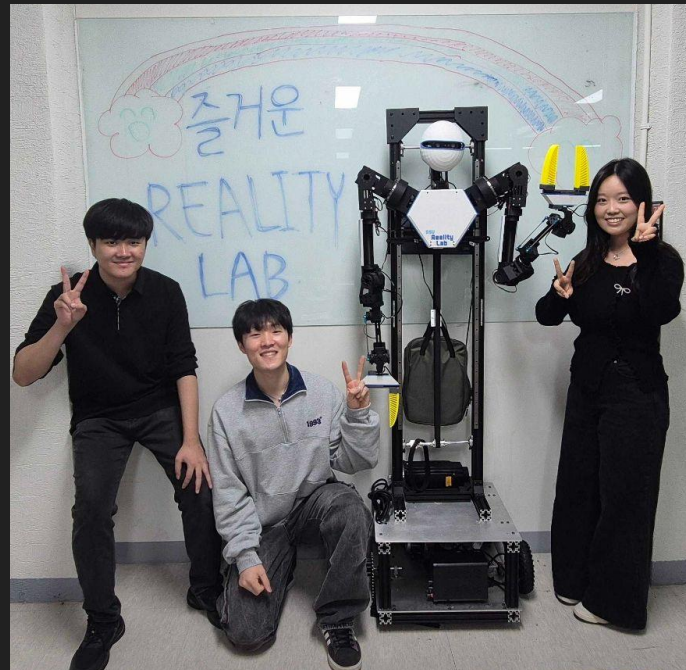
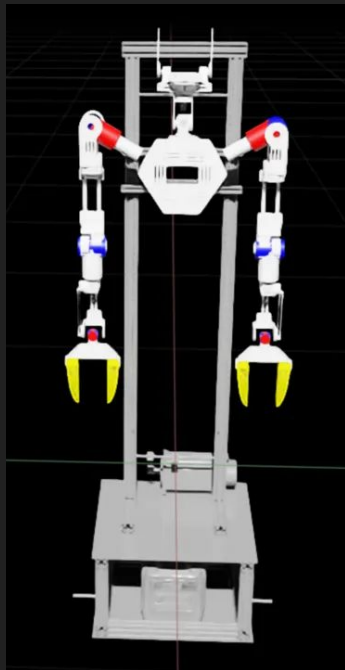
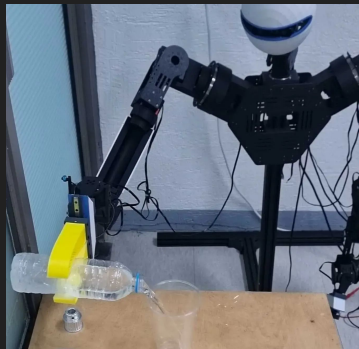


Appendix 2: VLA Guidance Method Research



Appendix 3: Humanoid Robot Project

: Integrated **RX1 upper body with a custom vertical mobile base**; Hardware assembly completed.
Currently focusing on mobile locomotion control and simulation environment setup.

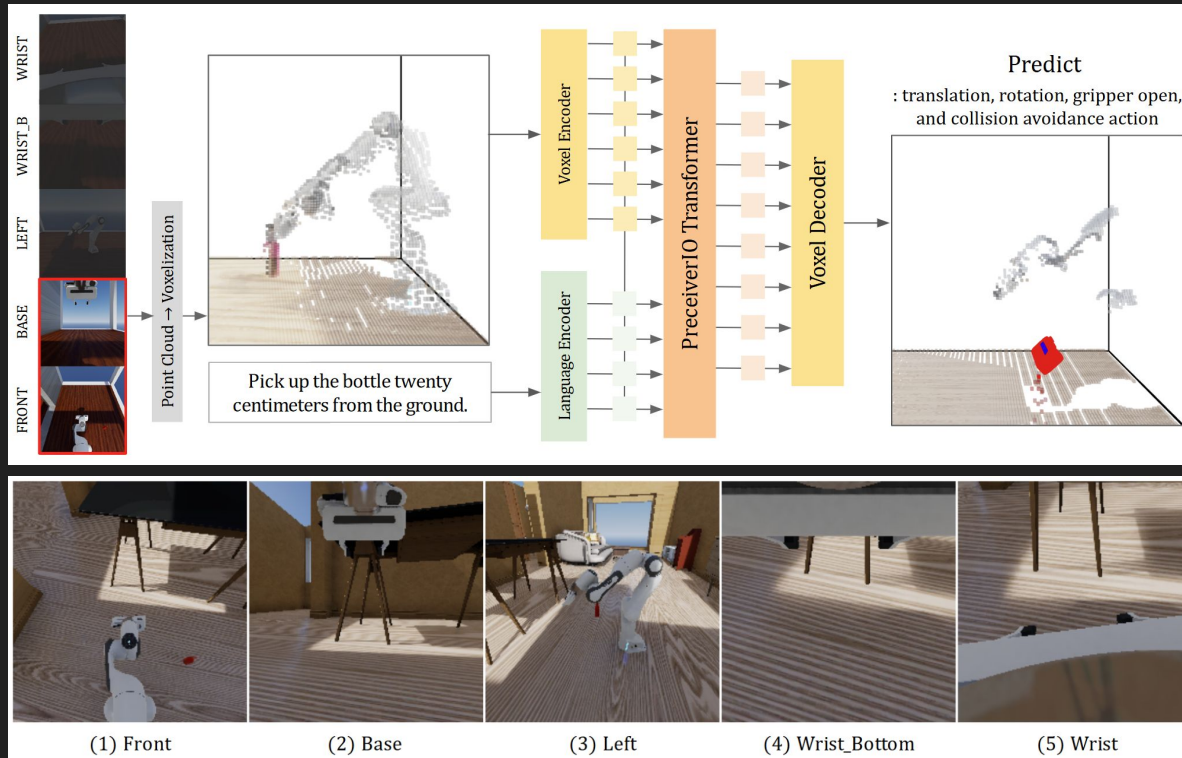


Appendix 4: Gyeonggi Youth Gap Year Program

: Development of an Open-Source Based **Small-Scale Humanoid**. (Grand Prize)



: This study shows that, for PerAct, **the number and placement of observation viewpoints strongly influence policy performance**, and that a small set of key views can match a full 5-view setup.



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Viewport	Success Rate (avg)
5-view	0.90
front	0.90
front + base	0.75
front + wrist	0.75
front + base + wrist	0.75

: **Policy performance** when the same viewpoint configuration is used for both training and evaluation.

Viewport	Novel Object	Novel Scene	Novel State	Any State
5-view	0.75	0.80	0.00	0.40
front	0.80	0.90	0.00	0.45
front + base	0.75	0.85	0.00	0.50
front + wrist_b	0.75	0.85	0.0	0.4
front + base + wrist_b	0.8	0.75	0.0	0.35

: **Generalization performance** of each policy trained under different viewpoint configurations.